PHYS4055 Atomic Physics and Quantum Optics

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1 Assignment 1

1.1 Question 1

1.1.1 Part a

We are given that the state of the system is

$$|\psi\rangle = \sqrt{1-\lambda^2}\sum_{n=0}^{\infty}\lambda^n|n\rangle_s|n\rangle_i$$

We begin by writing the density matrix of the complete signal–idler system:

$$\rho_{si} = |\psi\rangle\langle\psi|$$

Substituting the expression for $|\psi\rangle$, we obtain

$$\rho_{si} = (1-\lambda^2)\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\lambda^{n+m}|n\rangle_s|n\rangle_i\langle m|_s\langle m|_i$$

To obtain the reduced density matrix of the signal mode, we trace out the idler mode:

$$\rho_s = \mathrm{Tr}_i(\rho_{si})$$

The partial trace over the idler mode acts according to

$$\mathrm{Tr}_i\left(|n\rangle_s|n\rangle_i\langle m|_s\langle m|_i\right) = |n\rangle_s\langle m|_s\,\mathrm{Tr}\left(|n\rangle_i\langle m|_i\right)$$

Since

$$\mathrm{Tr}\left(|n\rangle_i\langle m|_i\right) = \langle m|n\rangle_i = \delta_{mn}$$

the only terms that remain are those for which $m = n$. Therefore, the reduced density matrix of the signal mode is

$$\rho_s = (1-\lambda^2)\sum_{n=0}^{\infty}\lambda^{2n}|n\rangle_s\langle n|_s$$

Due to the symmetry between the two modes, tracing out the signal mode gives the reduced density matrix of the idler mode:

$$\rho_i = (1-\lambda^2)\sum_{n=0}^{\infty} \lambda^{2n} |n\rangle_i \langle n|_i$$

We can now compare these expressions with the density matrix of a thermal state:

$$\rho_{\text{th}} = \frac{1}{1+\bar{n}} \sum_{n=0}^{\infty} \left(\frac{\bar{n}}{1+\bar{n}}\right)^n |n\rangle\langle n|$$

where $\bar{n}$ is the mean photon number.

By setting

$$\lambda^2 = \frac{\bar{n}}{1+\bar{n}}$$

we also have

$$1-\lambda^2 = \frac{1}{1+\bar{n}}$$

Therefore, the reduced density matrices ρ_s and ρ_i have exactly the form of a thermal state. Hence, both the signal and idler modes individually are thermal states.

The mean number of photons can now be calculated from the above expression as follows:

$$\lambda^2 = \frac{\bar{n}}{1+\bar{n}}$$

$$\lambda^2 + \lambda^2\bar{n} = \bar{n}$$

$$\lambda^2 = \bar{n} - \lambda^2\bar{n}$$

Therefore,

$$\boxed{\bar{n} = \frac{\lambda^2}{1-\lambda^2}}$$

Finally, since

$$\lambda = \tanh r,$$

we have

$$\bar{n} = \frac{\tanh^2 r}{1-\tanh^2 r}.$$

Using

$$1 - \tanh^2 r = \frac{1}{\cosh^2 r},$$

this becomes

$$\boxed{\bar{n} = \sinh^2 r}.$$

1.1.2 Part b

For the two-mode squeezed vacuum state,

$$|\Psi\rangle = \sqrt{1-\lambda^2} \sum_{n=0}^{\infty} \lambda^n |n\rangle_s |n\rangle_i,$$

the probability of finding n photons in both the signal and idler modes is

$$P(n) = \langle \psi | \psi \rangle = (1-\lambda^2)\lambda^{2n}.$$

Since photons are produced in pairs, we have

$$\hat{n}_s |n\rangle_s |n\rangle_i = n |n\rangle_s |n\rangle_i$$

and

$$\hat{n}_i |n\rangle_s |n\rangle_i = n |n\rangle_s |n\rangle_i.$$

Therefore,

$$\hat{n}_s \hat{n}_i |n\rangle_s |n\rangle_i = n^2 |n\rangle_s |n\rangle_i.$$

Hence,

$$\langle \hat{n}_s \hat{n}_i \rangle = (1-\lambda^2) \sum_{n=0}^{\infty} n^2 \lambda^{2n}.$$

Using the identity

$$\sum_{n=0}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}, \qquad |x| < 1,$$

with $x = \lambda^2$, we obtain

$$\begin{aligned} \langle \hat{n}_s \hat{n}_i \rangle &= (1-\lambda^2) \frac{\lambda^2(1+\lambda^2)}{(1-\lambda^2)^3} \\ &= \frac{\lambda^2(1+\lambda^2)}{(1-\lambda^2)^2}. \end{aligned}$$

From part (a),

$$\langle \hat{n}_s \rangle = \langle \hat{n}_i \rangle = \bar{n} = \frac{\lambda^2}{1-\lambda^2}.$$

The cross-correlation function is therefore

$$g_{si}^{(2)}(0) = \frac{\langle \hat{n}_s \hat{n}_i \rangle}{\langle \hat{n}_s \rangle \langle \hat{n}_i \rangle}.$$

Substituting the expressions above gives

$$\begin{aligned} g_{si}^{(2)}(0) &= \frac{\dfrac{\lambda^2(1+\lambda^2)}{(1-\lambda^2)^2}}{\left(\dfrac{\lambda^2}{1-\lambda^2}\right)^2} \\ &= \frac{1+\lambda^2}{\lambda^2}. \end{aligned}$$

Therefore,

$$\boxed{g_{si}^{(2)}(0) = \frac{1+\lambda^2}{\lambda^2}}.$$

Since

$$\lambda^2 = \frac{\bar{n}}{1+\bar{n}},$$

the result can also be written as

$$\boxed{g_{si}^{(2)}(0) = 2 + \frac{1}{\bar{n}}}$$

1.1.3 Part c

In the low-gain limit, $r \to 0$, we have

$$\lambda = \tanh r \approx r \ll 1.$$

The two-mode squeezed vacuum state can therefore be approximated as

$$|\Psi\rangle = \sqrt{1-\lambda^2}\left(|0\rangle_s|0\rangle_i + \lambda|1\rangle_s|1\rangle_i + \lambda^2|2\rangle_s|2\rangle_i + \cdots\right)$$

or, to first order in λ,

$$|\Psi\rangle \approx |0\rangle_s|0\rangle_i + \lambda|1\rangle_s|1\rangle_i.$$

The probability of producing two or more photon pairs is of higher order in λ, and is therefore negligible in the low-gain limit. Hence, the detection of one idler photon "heralds" the presence of a single photon in the signal mode.

1.2 Question 2

1.2.1 Part A

Start by writing the input state as a number state generated from the vacuum:

$$|n\rangle_A = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle_A$$

Therefore, the two-mode input state can be written as

$$|n\rangle_A|0\rangle_B = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle_A|0\rangle_B$$

Given the relation between the output and input modes,

$$\begin{pmatrix}\hat{c}\\ \hat{d}\end{pmatrix} = \begin{pmatrix}\tau & -\varrho\\ \varrho & \tau\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{b}\end{pmatrix}$$

we can use the fact that the beam-splitter matrix is real and orthogonal, so that

$$B^{-1} = B^T.$$

Hence,

$$\begin{pmatrix}\hat{a}\\ \hat{b}\end{pmatrix} = \begin{pmatrix}\tau & \varrho\\ -\varrho & \tau\end{pmatrix}\begin{pmatrix}\hat{c}\\ \hat{d}\end{pmatrix}.$$

Therefore,

$$\hat{a} = \tau\hat{c} + \varrho\hat{d},$$

and taking the Hermitian conjugate gives

$$\hat{a}^\dagger = \tau\hat{c}^\dagger + \varrho\hat{d}^\dagger.$$

We can now write the output state in terms of the output creation operators and the vacuum state:

$$|\psi_{\text{out}}\rangle = \frac{1}{\sqrt{n!}}\left(\tau\hat{c}^\dagger + \varrho\hat{d}^\dagger\right)^n|0\rangle_C|0\rangle_D.$$

As the creation operators of the two different modes commute,

$$[\hat{c}^\dagger, \hat{d}^\dagger] = 0,$$

this expression can be expanded using the binomial theorem:

$$|\psi_{\text{out}}\rangle = \frac{1}{\sqrt{n!}}\sum_{k=0}^{n}\binom{n}{k}\tau^k\varrho^{n-k}(\hat{c}^\dagger)^k(\hat{d}^\dagger)^{n-k}|0\rangle_C|0\rangle_D.$$

The creation operators acting on the vacuum give the corresponding number states:

$$(\hat{c}^\dagger)^k|0\rangle_C = \sqrt{k!}\,|k\rangle_C,$$

and

$$(\hat{d}^\dagger)^{n-k}|0\rangle_D = \sqrt{(n-k)!}\,|n-k\rangle_D.$$

Thus,

$$(\hat{c}^\dagger)^k(\hat{d}^\dagger)^{n-k}|0\rangle_C|0\rangle_D = \sqrt{k!(n-k)!}|k\rangle_C|n-k\rangle_D.$$

Substituting this into the previous expression and combining the factorials gives

$$|\psi_{\text{out}}\rangle = \sum_{k=0}^{n}\binom{n}{k}\sqrt{\frac{k!(n-k)!}{n!}}\tau^k\varrho^{n-k}|k\rangle_C|n-k\rangle_D.$$

Therefore,

$$\boxed{|n\rangle_A|0\rangle_B \to \sum_{k=0}^{n}\sqrt{\frac{n!}{k!(n-k)!}}\tau^k\varrho^{n-k}|k\rangle_C|n-k\rangle_D}$$

1.2.2 Part B

The coherent state entering port A can be expanded in the number basis as

$$|\alpha\rangle_A = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle_A.$$

Since the input to port B is the vacuum state, the two-mode input state is

$$|\psi_{\text{in}}\rangle = |\alpha\rangle_A|0\rangle_B = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle_A|0\rangle_B.$$

For the 50:50 beam splitter,

$$\hat{c} = \frac{\hat{a}+\hat{b}}{\sqrt{2}}, \qquad \hat{d} = \frac{\hat{a}-\hat{b}}{\sqrt{2}}.$$

These relations can be written in matrix form as

$$\begin{pmatrix}\hat{c}\\ \hat{d}\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{b}\end{pmatrix}.$$

Therefore, for this 50:50 beam splitter, all elements of the beam-splitter matrix have magnitude $1/\sqrt{2}$. Hence, the transmission and reflection probabilities are both

$$T = R = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}.$$

Thus, when applying the result from part (a), each factor of the transmission or reflection amplitude contributes a factor of $1/\sqrt{2}$.

Using the result from part (a),

$$|n\rangle_A|0\rangle_B \to \sum_{k=0}^{n} \sqrt{\frac{n!}{k!(n-k)!}} \left(\frac{1}{\sqrt{2}}\right)^n |k\rangle_C|n-k\rangle_D.$$

Substituting this into the coherent-state expansion gives

$$|\psi_{\text{out}}\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \sum_{k=0}^{n} \sqrt{\frac{n!}{k!(n-k)!}} \left(\frac{1}{\sqrt{2}}\right)^n |k\rangle_C|n-k\rangle_D.$$

Cancelling the factors of $\sqrt{n!}$, we obtain

$$|\psi_{\text{out}}\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{\left(\alpha/\sqrt{2}\right)^n}{\sqrt{k!(n-k)!}} |k\rangle_C|n-k\rangle_D.$$

Now define

$$m = n - k,$$

so that

$$n = k + m.$$

The double sum can therefore be rewritten as

$$|\psi_{\text{out}}\rangle = e^{-|\alpha|^2/2} \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \frac{\left(\alpha/\sqrt{2}\right)^{k+m}}{\sqrt{k!m!}} |k\rangle_C|m\rangle_D.$$

Since

$$\left(\frac{\alpha}{\sqrt{2}}\right)^{k+m} = \left(\frac{\alpha}{\sqrt{2}}\right)^k \left(\frac{\alpha}{\sqrt{2}}\right)^m,$$

and

$$e^{-|\alpha|^2/2} = e^{-|\alpha|^2/4} e^{-|\alpha|^2/4},$$

the state factorises as

$$|\psi_{\text{out}}\rangle = \left[e^{-|\alpha|^2/4} \sum_{k=0}^{\infty} \frac{\left(\alpha/\sqrt{2}\right)^k}{\sqrt{k!}} |k\rangle_C \right] \otimes \left[e^{-|\alpha|^2/4} \sum_{m=0}^{\infty} \frac{\left(\alpha/\sqrt{2}\right)^m}{\sqrt{m!}} |m\rangle_D \right].$$

Recognising each factor as a coherent state,

$$\left| \frac{\alpha}{\sqrt{2}} \right\rangle = e^{-|\alpha|^2/4} \sum_{n=0}^{\infty} \frac{\left(\alpha/\sqrt{2}\right)^n}{\sqrt{n!}} |n\rangle,$$

we find

$$\boxed{|\psi_{\text{out}}\rangle = \left| \frac{\alpha}{\sqrt{2}} \right\rangle_C \otimes \left| \frac{\alpha}{\sqrt{2}} \right\rangle_D}.$$

Therefore, the output of the 50:50 beam splitter is a product of coherent states.

1.2.3 Part C

From part (b), the output state in mode C is

$$\left| \frac{\alpha}{\sqrt{2}} \right\rangle_C.$$

Using the coherent-state expansion,

$$\left| \frac{\alpha}{\sqrt{2}} \right\rangle_C = e^{-|\alpha|^2/4} \sum_{n=0}^{\infty} \frac{\left(\alpha/\sqrt{2}\right)^n}{\sqrt{n!}} |n\rangle_C.$$

Therefore, the probability of detecting n photons in mode C is

$$P(n) = \left| e^{-|\alpha|^2/4} \frac{\left(\alpha/\sqrt{2}\right)^n}{\sqrt{n!}} \right|^2.$$

Hence,

$$\boxed{P(n) = e^{-|\alpha|^2/2} \frac{\left(|\alpha|^2/2\right)^n}{n!}}.$$

This is a Poisson distribution with mean photon number

$$\bar{n}_C = \frac{|\alpha|^2}{2}.$$

Therefore, the detector registers Poissonian photon-counting statistics.

2 Assignment 2

2.1 Question 1

2.1.1 Part A

(a) Bell-basis orthogonality

The computational basis states are orthonormal, so

$$\langle 0|0\rangle = 1, \qquad \langle 1|1\rangle = 1, \qquad \langle 0|1\rangle = \langle 1|0\rangle = 0.$$

The four Bell states are

$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle \pm |11\rangle\right), \qquad |\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle \pm |10\rangle\right).$$

We now calculate the inner products between each distinct pair of Bell states. First,

$$\begin{aligned}
\langle\Phi^+|\Phi^-\rangle &= \frac{1}{2}\left(\langle 00| + \langle 11|\right)\left(|00\rangle - |11\rangle\right) \\
&= \frac{1}{2}\left(\langle 00|00\rangle - \langle 00|11\rangle + \langle 11|00\rangle - \langle 11|11\rangle\right) \\
&= \frac{1}{2}(1 - 0 + 0 - 1) = 0.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\langle\Phi^+|\Psi^+\rangle &= \frac{1}{2}\left(\langle 00| + \langle 11|\right)\left(|01\rangle + |10\rangle\right) \\
&= \frac{1}{2}\left(\langle 00|01\rangle + \langle 00|10\rangle + \langle 11|01\rangle + \langle 11|10\rangle\right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
\langle\Phi^+|\Psi^-\rangle &= \frac{1}{2}\left(\langle 00| + \langle 11|\right)\left(|01\rangle - |10\rangle\right) \\
&= \frac{1}{2}\left(\langle 00|01\rangle - \langle 00|10\rangle + \langle 11|01\rangle - \langle 11|10\rangle\right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
\langle\Phi^-|\Psi^+\rangle &= \frac{1}{2}\left(\langle 00| - \langle 11|\right)\left(|01\rangle + |10\rangle\right) \\
&= \frac{1}{2}\left(\langle 00|01\rangle + \langle 00|10\rangle - \langle 11|01\rangle - \langle 11|10\rangle\right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
\langle\Phi^-|\Psi^-\rangle &= \frac{1}{2}\left(\langle 00| - \langle 11|\right)\left(|01\rangle - |10\rangle\right) \\
&= \frac{1}{2}\left(\langle 00|01\rangle - \langle 00|10\rangle - \langle 11|01\rangle + \langle 11|10\rangle\right) \\
&= 0.
\end{aligned}$$

Finally,

$$\begin{aligned}\langle\Psi^+|\Psi^-\rangle &= \frac{1}{2}\left(\langle 01| + \langle 10|\right)\left(|01\rangle - |10\rangle\right)\\ &= \frac{1}{2}\left(\langle 01|01\rangle - \langle 01|10\rangle + \langle 10|01\rangle - \langle 10|10\rangle\right)\\ &= \frac{1}{2}(1 - 0 + 0 - 1) = 0.\end{aligned}$$

Thus, every pair of distinct Bell states has zero inner product. Since each Bell state is also normalised,

$$\langle\Phi^\pm|\Phi^\pm\rangle = \langle\Psi^\pm|\Psi^\pm\rangle = 1,$$

the four Bell states form an orthonormal basis.

2.1.2 Part B

(b) Local state of one photon

The two-photon state is

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right).$$

The density operator is therefore

$$\begin{aligned}\rho_{AB} &= |\Phi^+\rangle\langle\Phi^+|\\ &= \frac{1}{2}\left(|00\rangle + |11\rangle\right)\left(\langle 00| + \langle 11|\right)\\ &= \frac{1}{2}\left(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|\right).\end{aligned}$$

The reduced density operator for photon A is obtained by tracing over photon B:

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \sum_{n=0}^{1} {}_B\langle n|\rho_{AB}|n\rangle_B.$$

Hence,

$$\begin{aligned}\rho_A &= {}_B\langle 0|\rho_{AB}|0\rangle_B + {}_B\langle 1|\rho_{AB}|1\rangle_B\\ &= \frac{1}{2}\Big(|0\rangle_A\langle 0|_A\langle 0|0\rangle_B\langle 0|0\rangle_B + |0\rangle_A\langle 1|_A\langle 0|0\rangle_B\langle 1|0\rangle_B\\ &\quad + |1\rangle_A\langle 0|_A\langle 0|1\rangle_B\langle 0|0\rangle_B + |1\rangle_A\langle 1|_A\langle 0|1\rangle_B\langle 1|0\rangle_B\Big)\\ &\quad + \frac{1}{2}\Big(|0\rangle_A\langle 0|_A\langle 1|0\rangle_B\langle 0|1\rangle_B + |0\rangle_A\langle 1|_A\langle 1|0\rangle_B\langle 1|1\rangle_B\\ &\quad + |1\rangle_A\langle 0|_A\langle 1|1\rangle_B\langle 0|1\rangle_B + |1\rangle_A\langle 1|_A\langle 1|1\rangle_B\langle 1|1\rangle_B\Big).\end{aligned}$$

Using

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \qquad \langle 0|1\rangle = \langle 1|0\rangle = 0,$$

this reduces to

$$\rho_A = \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right).$$

Therefore,

$$\boxed{\rho_A = \frac{1}{2}I}$$

or, using the polarisation encoding,

$$\rho_A = \frac{1}{2}\left(|H\rangle\langle H| + |V\rangle\langle V|\right).$$

Thus, a measurement of photon A alone in the $\{|H\rangle, |V\rangle\}$ basis gives

$$P(H) = P(V) = \frac{1}{2}.$$

Although the two photons are in a pure entangled state, photon A by itself is in a maximally mixed state.

2.1.3 Part C

(c) Correlations in two bases

From

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \qquad |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}},$$

we can write

$$|H\rangle = \frac{|D\rangle + |A\rangle}{\sqrt{2}}, \qquad |V\rangle = \frac{|D\rangle - |A\rangle}{\sqrt{2}}.$$

Starting with

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|HH\rangle + |VV\rangle\right),$$

substituting for $|H\rangle$ and $|V\rangle$ gives

$$\begin{aligned}
|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\left[\left(\frac{|D\rangle + |A\rangle}{\sqrt{2}}\right)\left(\frac{|D\rangle + |A\rangle}{\sqrt{2}}\right) + \left(\frac{|D\rangle - |A\rangle}{\sqrt{2}}\right)\left(\frac{|D\rangle - |A\rangle}{\sqrt{2}}\right)\right] \\
&= \frac{1}{2\sqrt{2}}\left[|DD\rangle + |DA\rangle + |AD\rangle + |AA\rangle + |DD\rangle - |DA\rangle - |AD\rangle + |AA\rangle\right] \\
&= \frac{1}{\sqrt{2}}\left(|DD\rangle + |AA\rangle\right).
\end{aligned}$$

(i) In the $\{|H\rangle, |V\rangle\}$ basis,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|HH\rangle + |VV\rangle\right).$$

Therefore, either both photons are measured in the H state or both are measured in the V state. The outcomes are perfectly correlated.

(ii) In the $\{|D\rangle, |A\rangle\}$ basis,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} \left(|DD\rangle + |AA\rangle \right).$$

Therefore, either both photons are measured in the D state or both are measured in the A state. The outcomes are again perfectly correlated.

2.2 Question 2

2.2.1 Part A

The initial shared Bell state is

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right).$$

Alice applies one of the operators

$$I, \qquad X, \qquad Z, \qquad XZ$$

to her photon only.

For the identity operator,

$$\begin{aligned}(I \otimes I)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \\ &= |\Phi^+\rangle.\end{aligned}$$

For the X operator,

$$\begin{aligned}(X \otimes I)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\left[(X \otimes I)|00\rangle + (X \otimes I)|11\rangle\right] \\ &= \frac{1}{\sqrt{2}}\left(|10\rangle + |01\rangle\right) \\ &= |\Psi^+\rangle.\end{aligned}$$

For the Z operator,

$$\begin{aligned}(Z \otimes I)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\left[(Z \otimes I)|00\rangle + (Z \otimes I)|11\rangle\right] \\ &= \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right) \\ &= |\Phi^-\rangle.\end{aligned}$$

Finally, for the XZ operator,

$$XZ|0\rangle = |1\rangle, \qquad XZ|1\rangle = -|0\rangle,$$

so

$$\begin{aligned}(XZ \otimes I)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\left[(XZ \otimes I)|00\rangle + (XZ \otimes I)|11\rangle\right] \\ &= \frac{1}{\sqrt{2}}\left(|10\rangle - |01\rangle\right) \\ &= -|\Psi^-\rangle.\end{aligned}$$

Since an overall phase of -1 has no physical significance,

$$(XZ \otimes I)|\Phi^+\rangle \sim |\Psi^-\rangle.$$

A consistent encoding is therefore

Classical message	Alice's operation	Resulting Bell state
00	I	$\|\Phi^+\rangle$
01	X	$\|\Psi^+\rangle$
10	Z	$\|\Phi^-\rangle$
11	XZ	$\|\Psi^-\rangle$

2.2.2 Part B

After Alice applies her chosen operation, she sends her photon to Bob. Bob now has both photons and performs a Bell-state measurement.

From part (a), the four possible messages correspond to four different Bell states:

$$00 \to |\Phi^+\rangle, \qquad 01 \to |\Psi^+\rangle, \qquad 10 \to |\Phi^-\rangle, \qquad 11 \to |\Psi^-\rangle.$$

An ideal Bell-state measurement can distinguish these four orthogonal states, so the measurement outcome tells Bob which operation Alice applied and therefore which two-bit message was sent.

Thus, although Alice physically sends only one photon, the information is encoded in the joint entangled two-photon state, allowing Bob to recover 2 classical bits.

2.2.3 Part C

The four Bell states may be written in terms of the input-mode creation operators as

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}\left(\hat{a}_H^\dagger \hat{b}_H^\dagger \pm \hat{a}_V^\dagger \hat{b}_V^\dagger\right)|0\rangle,$$

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}\left(\hat{a}_H^\dagger \hat{b}_V^\dagger \pm \hat{a}_V^\dagger \hat{b}_H^\dagger\right)|0\rangle.$$

Using the beam-splitter transformations

$$\hat{a}_\lambda^\dagger \to \frac{\hat{c}_\lambda^\dagger + \hat{d}_\lambda^\dagger}{\sqrt{2}}, \qquad \hat{b}_\lambda^\dagger \to \frac{\hat{c}_\lambda^\dagger - \hat{d}_\lambda^\dagger}{\sqrt{2}},$$

we obtain the following.

For $|\Phi^+\rangle$,

$$\begin{aligned}|\Phi^+\rangle &\to \frac{1}{2\sqrt{2}}\Big[(\hat{c}_H^\dagger + \hat{d}_H^\dagger)(\hat{c}_H^\dagger - \hat{d}_H^\dagger) + (\hat{c}_V^\dagger + \hat{d}_V^\dagger)(\hat{c}_V^\dagger - \hat{d}_V^\dagger)\Big]|0\rangle \\ &= \frac{1}{2\sqrt{2}}\Big[(\hat{c}_H^\dagger)^2 - (\hat{d}_H^\dagger)^2 + (\hat{c}_V^\dagger)^2 - (\hat{d}_V^\dagger)^2\Big]|0\rangle.\end{aligned}$$

For $|\Phi^-\rangle$,

$$|\Phi^-\rangle \to \frac{1}{2\sqrt{2}}\Big[(\hat{c}_H^\dagger + \hat{d}_H^\dagger)(\hat{c}_H^\dagger - \hat{d}_H^\dagger) - (\hat{c}_V^\dagger + \hat{d}_V^\dagger)(\hat{c}_V^\dagger - \hat{d}_V^\dagger)\Big]|0\rangle$$
$$= \frac{1}{2\sqrt{2}}\Big[(\hat{c}_H^\dagger)^2 - (\hat{d}_H^\dagger)^2 - (\hat{c}_V^\dagger)^2 + (\hat{d}_V^\dagger)^2\Big]|0\rangle.$$

For $|\Psi^+\rangle$,

$$|\Psi^+\rangle \to \frac{1}{2\sqrt{2}}\Big[(\hat{c}_H^\dagger + \hat{d}_H^\dagger)(\hat{c}_V^\dagger - \hat{d}_V^\dagger) + (\hat{c}_V^\dagger + \hat{d}_V^\dagger)(\hat{c}_H^\dagger - \hat{d}_H^\dagger)\Big]|0\rangle$$
$$= \frac{1}{\sqrt{2}}\left(\hat{c}_H^\dagger \hat{c}_V^\dagger - \hat{d}_H^\dagger \hat{d}_V^\dagger\right)|0\rangle.$$

Finally, for $|\Psi^-\rangle$,

$$|\Psi^-\rangle \to \frac{1}{2\sqrt{2}}\Big[(\hat{c}_H^\dagger + \hat{d}_H^\dagger)(\hat{c}_V^\dagger - \hat{d}_V^\dagger) - (\hat{c}_V^\dagger + \hat{d}_V^\dagger)(\hat{c}_H^\dagger - \hat{d}_H^\dagger)\Big]|0\rangle$$
$$= \frac{1}{\sqrt{2}}\left(\hat{d}_H^\dagger \hat{c}_V^\dagger - \hat{c}_H^\dagger \hat{d}_V^\dagger\right)|0\rangle.$$

The possible detector outcomes are therefore

Bell state	Possible output pattern
$\|\Phi^+\rangle$	HH or VV together in c or d
$\|\Phi^-\rangle$	HH or VV together in c or d
$\|\Psi^+\rangle$	HV together in c or d
$\|\Psi^-\rangle$	H and V in different output ports

Thus,

$$\boxed{|\Psi^+\rangle \text{ and } |\Psi^-\rangle \text{ can be distinguished unambiguously.}}$$

However, $|\Phi^+\rangle$ and $|\Phi^-\rangle$ produce the same photon-number and polarisation patterns and differ only in relative phase. Therefore,

$$\boxed{|\Phi^+\rangle \text{ and } |\Phi^-\rangle \text{ remain indistinguishable.}}$$

2.2.4 Part D

From part (c), only the states

$$|\Psi^+\rangle \qquad \text{and} \qquad |\Psi^-\rangle$$

can be identified unambiguously by the measurements.

Since the four messages are equally likely,

$$P(\text{success}) = \frac{2}{4} = \frac{1}{2}.$$

Therefore,

$$\boxed{P(\text{success}) = 50\%}.$$

Because $|\Phi^+\rangle$ and $|\Phi^-\rangle$ cannot be distinguished, Bob cannot determine Alice's message with certainty on every run. Hence this measurement alone does not allow deterministic dense coding.

2.3 Question 3

2.3.1 Part A

Alice randomly chooses a bit value and either the Z or X polarisation basis.

Basis	Bit value	State sent
Z	0	$\|H\rangle$
Z	1	$\|V\rangle$
X	0	$\|D\rangle$
X	1	$\|A\rangle$

where

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \qquad |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}.$$

2.3.2 Part B

Alice and Bob retain a result only when they have chosen the same basis. Since each independently chooses Z or X with equal probability,

$$P(\text{same basis}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Therefore, the expected fraction of Bob's detections that remain in the sifted key is

$$\boxed{\frac{1}{2}}.$$

If Alice sends N photons, the expected number detected by Bob is

$$\boxed{N_{\text{det}} = N\eta_{\text{ch}}\eta_d}.$$

Since only half of these detections are retained after basis sifting, the expected sifted-key length is

$$\boxed{N_{\text{sift}} = \frac{1}{2}N\eta_{\text{ch}}\eta_d}.$$

2.3.3 Part C

Since Eve randomly chooses either the Z or X basis,

$$P(\text{Eve uses Alice's basis}) = \frac{1}{2}, \qquad P(\text{Eve uses a different basis}) = \frac{1}{2}.$$

If Eve uses the same basis as Alice, she obtains the correct result and resends the correct state, so

$$P(\text{error} \mid \text{same basis}) = 0.$$

If Eve uses the wrong basis, her measurement result is random. When Bob then measures in Alice's basis, he has a probability

$$P(\text{error} \mid \text{different basis}) = \frac{1}{2}$$

of obtaining the wrong bit.

Therefore,

$$\begin{aligned} Q &= P(\text{same basis})P(\text{error} \mid \text{same basis}) \\ &\quad + P(\text{different basis})P(\text{error} \mid \text{different basis}) \\ &= \frac{1}{2}(0) + \frac{1}{2}\left(\frac{1}{2}\right) \\ &= \frac{1}{4}. \end{aligned}$$

Hence,

$$\boxed{Q = 25\%}.$$

2.3.4 Part D

From part (c),

$$Q = \frac{1}{4}.$$

(i) For $m = 20$,

$$P(\text{at least one error}) = 1 - (1 - Q)^{20} = 1 - \left(\frac{3}{4}\right)^{20}.$$

Therefore,

$$P(\text{at least one error}) \approx 0.9968,$$

so

$$\boxed{P(\text{at least one error}) \approx 99.7\%}.$$

(ii) The 20 bits used to estimate the QBER are publicly revealed, so Eve may also know their values. They therefore cannot be included in the final secret key.

(iii) **Error correction** is used to remove disagreements between Alice's and Bob's sifted keys so that they obtain identical bit strings.

Privacy amplification compresses the corrected key into a shorter key, reducing any partial information that Eve may have gained.

3 Assignment 3

1. Nuclear-size effects

a)

For a point nucleus,

$$V_0(r) = -\frac{Z}{r},$$

while for a uniformly charged nucleus, for $r < R$,

$$V(r) = -\frac{Z}{R}\left[\frac{3}{2} - \frac{1}{2}\left(\frac{r}{R}\right)^2\right].$$

Hence the perturbation is

$$\delta V(r) = V(r) - V_0(r) = Z\left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3}\right), \qquad r < R,$$

and $\delta V(r) = 0$ for $r \geq R$.

Using first-order perturbation theory,

$$\Delta E_{\text{fns}} = \langle\psi|\delta V|\psi\rangle = \int \psi^*(\mathbf{r})\,\delta V(r)\,\psi(\mathbf{r})\,d^3r.$$

For an ns state, the nuclear radius is much smaller than the characteristic atomic length scale, so within the nucleus

$$|\psi_{ns}(r)|^2 \simeq |\psi_{ns}(0)|^2.$$

From

$$\psi_{nlm}(\mathbf{r}) = \frac{P_{nl}(r)}{r} Y_{lm}(\theta, \phi),$$

for $l = 0$ and $r \to 0$,

$$P_{n0}(r) \simeq N_{n0}\frac{2Zr}{n}, \qquad N_{n0} = \sqrt{\frac{Z}{n}}.$$

Hence

$$\frac{P_{n0}(r)}{r} \to \frac{2Z^{3/2}}{n^{3/2}}.$$

Since

$$Y_{00} = \frac{1}{\sqrt{4\pi}},$$

we obtain

$$\psi_{ns}(0) = \frac{Z^{3/2}}{\sqrt{\pi n^3}}, \qquad |\psi_{ns}(0)|^2 = \frac{Z^3}{\pi n^3}.$$

Therefore, using $d^3r = 4\pi r^2 dr$,

$$\Delta E_{\rm fns} \simeq 4\pi |\psi_{ns}(0)|^2 \int_0^R \delta V(r) r^2 \, dr.$$

Thus

$$\begin{aligned}
\Delta E_{\rm fns} &= 4\pi \frac{Z^3}{\pi n^3} Z \int_0^R \left(r - \frac{3r^2}{2R} + \frac{r^4}{2R^3} \right) dr \\
&= 4\frac{Z^4}{n^3} \left(\frac{R^2}{2} - \frac{R^2}{2} + \frac{R^2}{10} \right) \\
&= \frac{2Z^4 R^2}{5n^3}.
\end{aligned}$$

Hence the leading finite-nuclear-size correction for an s state is

$$\boxed{\Delta E_{\rm fns} = \frac{2Z^2}{5n^3} (ZR)^2}.$$

b)

The finite-nuclear-size correction was evaluated numerically using the hydrogen-like radial wavefunctions $P_{nl}(r)$ provided in the Mathematica notebooks. In atomic units, the first-order correction is

$$\Delta E_{\mathrm{fns}} = \int_0^R P_{nl}(r)^2 \, \delta V(r) \, dr,$$

where

$$\delta V(r) = Z \left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3} \right), \qquad r < R.$$

The following Mathematica code was used:

```
(* Finite nuclear-size corrections for H-like 209Bi.
   Atomic units are used throughout. *)

ClearAll["Global`*"];

(* Hydrogen-like radial function P_nl(r) *)
P[{n_, l_}, Z_, r_] := Block[{\[Rho] = 2 r/n},
  1/(n (2 l + 1)!) *
   Sqrt[Z (n + l)!/(n - l - 1)!] *
   (Z \[Rho])^(l + 1) *
   Exp[-Z \[Rho]/2] *
   Hypergeometric1F1[-n + l + 1, 2 l + 2, Z \[Rho]]
  ];

(* Nuclear parameters for 209Bi *)
Z = 83;
Rrms = 5.5211/52917.7210903;     (* fm -> a.u. *)
R = Sqrt[5/3] Rrms;

(* First-order finite-nuclear-size correction.
   x = r/R is used for numerical stability. *)
deltaE[n_, l_] := NIntegrate[
  Evaluate@Simplify[
    P[{n, l}, Z, R x]^2 *
     Z (1/x - 3/2 + x^2/2),
    Assumptions -> 0 < x <= 1
    ],
  {x, 0, 1},
  WorkingPrecision -> 40
  ];

(* Numerical corrections *)
dE1s = deltaE[1, 0];
dE2p = deltaE[2, 1];
dE3d = deltaE[3, 2];

```

```
(* Leading-order analytic result from part (a) *)
dE1sAnalytic = (2 Z^2/5) (Z R)^2;

Grid[
 {
  {"State", "Numerical Delta E_fns (a.u.)",
   "Analytic Delta E_fns (a.u.)"},
  {"1s", N[dE1s, 10], N[dE1sAnalytic, 10]},
  {"2p", N[dE2p, 10], "-"},
  {"3d", N[dE3d, 10], "-"}
 },
 Frame -> All
]
```

The resulting energy corrections are

State	$\Delta E_{\mathrm{fns}}^{\mathrm{num}}$ (a.u.)	$\Delta E_{\mathrm{fns}}^{\mathrm{analytic}}$ (a.u.)
$1s$	0.3412152609	0.344405
$2p$	9.54585×10^{-8}	–
$3d$	8.63359×10^{-15}	–

For the $1s$ state, the numerical value differs from the leading-order analytical result from part (a) by

$$\frac{|0.341215 - 0.344405|}{0.344405} \times 100\% \simeq 0.93\%.$$

Thus, the numerical and analytical results are in very good agreement. The small difference is expected because the analytical result keeps only the leading term in R and approximates the s-state wavefunction as constant across the nucleus, whereas the numerical calculation uses the full radial wavefunction.

The corrections for the $2p$ and $3d$ states are much smaller. This is because near the origin the hydrogenic radial wavefunction behaves as $\psi \propto r^l$, so states with $l > 0$ have very little probability density within the nucleus. Consequently the finite-nuclear-size effect is strongly suppressed for the p state and even more strongly for the d state.

c)

From the relativistic correction derived in class,

$$\Delta E_{\text{rel}} = -\left(\frac{1}{l+\frac{1}{2}} - \frac{3}{4n}\right)\frac{Z^4\alpha^2}{n^3}\,\text{Ry},$$

where Ry $= 1/2$ a.u. Using $Z = 83$ and $\alpha \simeq 1/137.036$ gives

State	ΔE_{rel} (a.u.)	ΔE_{fns} (a.u.)
$1s$	-1.58×10^3	3.41×10^{-1}
$2p$	-4.61×10^1	9.55×10^{-8}
$3d$	-7.02	8.63×10^{-15}

Thus the relativistic correction is much larger than the finite-nuclear-size correction for all three states. For the $1s$ state, the finite-size correction is still measurable on the atomic energy scale, but it is about four orders of magnitude smaller than the relativistic correction.

For the $2p$ and $3d$ states the finite-nuclear-size correction becomes extremely small. This is because states with $l > 0$ have wavefunctions that vanish at the origin, so very little probability density lies inside the nucleus. In contrast, the relativistic correction depends on the behaviour of the electron over the whole atomic wavefunction and remains significant.

The finite-nuclear-size correction is positive, since the finite nuclear charge distribution makes the potential less attractive near the origin, whereas the relativistic correction shown above is negative.

2. Electron configurations and terms

a)

For the configuration nd^2, both electrons have

$$l_1 = l_2 = 2, \qquad s_1 = s_2 = \frac{1}{2}.$$

In LS coupling,

$$L = |l_1 - l_2|, \ldots, l_1 + l_2 = 0, 1, 2, 3, 4,$$

corresponding to

$$S,\ P,\ D,\ F,\ G,$$

while

$$S = |s_1 - s_2|, \ldots, s_1 + s_2 = 0, 1.$$

Because the two electrons are identical fermions in the same d subshell, the Pauli principle restricts the allowed combinations. For equivalent d electrons,

L	orbital symmetry	allowed spin
$0, 2, 4$	symmetric	$S = 0$
$1, 3$	antisymmetric	$S = 1$

Hence the allowed LS terms are

$$^1S, \qquad ^1D, \qquad ^1G, \qquad ^3P, \qquad ^3F.$$

The possible values of J are

$$J = |L - S|, \ldots, L + S.$$

Therefore,

$$^1S_0, \qquad ^1D_2, \qquad ^1G_4,$$
$$^3P_0, \quad ^3P_1, \quad ^3P_2,$$

and

$$^3F_2, \quad ^3F_3, \quad ^3F_4.$$

Thus the complete set of terms is

$$\boxed{^1S_0,\ ^1D_2,\ ^1G_4,\ ^3P_{0,1,2},\ ^3F_{2,3,4}}.$$

b)

The expected ordering may be determined using Hund's rules.

The first Hund rule states that the term with the largest total spin S has the lowest energy. Hence the triplet terms ($S = 1$) are favoured over the singlet terms ($S = 0$), and the ground term must be one of

$$^3F, \qquad ^3P.$$

The second Hund rule states that, for terms with the same S, the term with the largest L has the lowest energy. Therefore, for the triplets,

$$^3F < ^3P,$$

since $L = 3$ for ^3F and $L = 1$ for ^3P.

Similarly, for the singlet terms,

$$^1G < ^1D < ^1S,$$

since

$$L = 4,\ 2,\ 0,$$

respectively.

Thus Hund's rules predict that the ground term is

$$\boxed{^3F}.$$

The rules used above do not, by themselves, determine the relative position of ^3P compared with each of the singlet terms; this must be determined from the detailed electron–electron interaction or from the observed spectrum.

For the fine-structure splitting within a term, the usual third Hund rule states that for a less-than-half-filled subshell, such as d^2, the smallest J has the lowest energy. Hence

$$^3F_2 < ^3F_3 < ^3F_4,$$

and

$$^3P_0 < ^3P_1 < ^3P_2.$$

c)

Titanium was chosen, with ground-state configuration

$$\mathrm{Ti} : [\mathrm{Ar}]\, 3d^2 4s^2,$$

so that the open subshell is $3d^2$ while the $4s^2$ subshell is filled.

From the NIST Atomic Spectra Database, the relevant levels associated with the $3d^2 4s^2$ configuration are

Level	E (cm^{-1})
$^3F_2$	0.000
$^3F_3$	170.1328
$^3F_4$	386.874
$^1D_2$	7255.355
$^3P_0$	8436.618
$^3P_1$	8492.422
$^3P_2$	8602.3441
$^1G_4$	12118.393
$^1S_0$	not listed

Thus, for the levels listed by NIST, the observed ordering is

$$\boxed{{}^3F_2 < {}^3F_3 < {}^3F_4 < {}^1D_2 < {}^3P_0 < {}^3P_1 < {}^3P_2 < {}^1G_4}.$$

There is good agreement with the Hund's-rule prediction for the ground term: $^3F_2$ is the ground state. The ordering within each triplet multiplet also agrees with the expectation for a less-than-half-filled d subshell,

$$^3F_2 < {}^3F_3 < {}^3F_4, \qquad {}^3P_0 < {}^3P_1 < {}^3P_2.$$

This illustrates that Hund's rules are most reliable for predicting the ground term and qualitative trends, rather than the complete ordering of all excited terms. The detailed level positions depend on the electron–electron interaction and other corrections not included in the simple LS-coupling argument.

d)

Consider the excited state $^1D_2$. From the NIST data,

$$\Delta E = 7255.355 \text{ cm}^{-1}.$$

Using

$$1 \text{ a.u.} = 1 \ E_h = 219474.63 \text{ cm}^{-1},$$

the excitation energy in atomic units is

$$\Delta E = \frac{7255.355}{219474.63} = 0.03306 \text{ a.u.}$$

Therefore,

$$\boxed{\Delta E(^1D_2) = 7255.355 \text{ cm}^{-1} = 0.03306 \text{ a.u.}}$$

Question 3. Atomic Polarisability

a)

For the ground state,

$$\Delta E_1 = \sum_m \frac{|\langle 1|H_E|m\rangle|^2}{E_1 - E_m}, \qquad H_E = \epsilon z$$

in atomic units.

Using the approximation

$$E_1 - E_m \approx E_1 - E_2,$$

gives

$$\Delta E_1 \approx \frac{\epsilon^2}{E_1 - E_2} \sum_m |\langle 1|z|m\rangle|^2.$$

Using closure,

$$\sum_m |m\rangle\langle m| = 1,$$

so

$$\sum_m |\langle 1|z|m\rangle|^2 = \langle 1|z^2|1\rangle.$$

For a hydrogen-like ion,

$$E_n = -\frac{Z^2}{2n^2},$$

hence

$$E_1 - E_2 = -\frac{Z^2}{2} + \frac{Z^2}{8} = -\frac{3Z^2}{8}.$$

For the spherically symmetric $1s$ state,

$$\langle r^2\rangle = \langle x^2\rangle + \langle y^2\rangle + \langle z^2\rangle$$

Hence,

$$\langle z^2\rangle = \frac{1}{3}\langle r^2\rangle = \frac{1}{Z^2}.$$

Therefore,

$$\Delta E_1 = \frac{\epsilon^2}{-3Z^2/8}\frac{1}{Z^2} = -\frac{8}{3Z^4}\epsilon^2.$$

Comparing with

$$\Delta E_1 = -\frac{1}{2}\alpha\epsilon^2,$$

we obtain

$$\boxed{\alpha = \frac{16}{3Z^4}}$$

in atomic units.

b)

For hydrogen, $Z = 1$, so from part (a)

$$\alpha = \frac{16}{3} a_B^3 = 5.33\, a_B^3.$$

Using

$$a_B = 5.29177 \times 10^{-9} \text{ cm},$$

gives

$$\alpha = \frac{16}{3}(5.29177 \times 10^{-9})^3 = 7.90 \times 10^{-25} \text{ cm}^3.$$

Therefore,

$$\boxed{\alpha \approx 5.33\, a_B^3 \approx 7.90 \times 10^{-25} \text{ cm}^3}.$$

The exact hydrogen $1s$ value is

$$\alpha_{\text{exact}} = \frac{9}{2} a_B^3 = 4.50\, a_B^3.$$

Thus, the approximation overestimates the exact value by

$$\frac{5.33 - 4.50}{4.50} \times 100\% \approx 18.5\%.$$

c)

For hydrogen-like rubidium, $Z = 37$, so from part (a)

$$\alpha = \frac{16}{3Z^4} a_B^3 = \frac{16}{3(37)^4} a_B^3 = 2.85 \times 10^{-6} a_B^3.$$

Using

$$a_B = 5.29177 \times 10^{-9} \text{ cm},$$

gives

$$\alpha = 2.85 \times 10^{-6}(5.29177 \times 10^{-9})^3 = 4.22 \times 10^{-31} \text{ cm}^3.$$

Therefore,

$$\boxed{\alpha \approx 2.85 \times 10^{-6} a_B^3 \approx 4.22 \times 10^{-31} \text{ cm}^3}.$$

This is vastly smaller than the ground-state polarisability of neutral rubidium,

$$\alpha_{\text{Rb}} \approx 319\, a_B^3.$$

For hydrogen-like rubidium, the single $1s$ electron experiences the full nuclear charge $Z = 37$ and is therefore very tightly bound and concentrated close to the nucleus. Its electron cloud is consequently very difficult to distort. In neutral rubidium, the outer $5s$ valence electron is strongly screened by the inner electrons and is much more weakly bound, making the atom far more polarisable.

Question 4. Lande Factor

a)

The magnetic interaction is

$$H_B = -\boldsymbol{\mu} \cdot \mathbf{B},$$

with

$$\boldsymbol{\mu} = -\mu_B(g_L\mathbf{L} + g_S\mathbf{S}).$$

Taking

$$g_L = 1, \qquad g_S = 2,$$

gives

$$\boldsymbol{\mu} = -\mu_B(\mathbf{L} + 2\mathbf{S}).$$

In LS coupling,

$$\mathbf{J} = \mathbf{L} + \mathbf{S},$$

and we define the effective magnetic moment by

$$\boldsymbol{\mu} = -g_J\mu_B\mathbf{J}.$$

Taking the scalar product with $\mathbf{J}$ and then the expectation value,

$$g_J\langle\mathbf{J}^2\rangle = \langle(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{J}\rangle\,.$$

Since

$$\mathbf{L} + 2\mathbf{S} = \mathbf{J} + \mathbf{S},$$

we have

$$g_J\langle\mathbf{J}^2\rangle = \langle\mathbf{J}^2\rangle + \langle\mathbf{S} \cdot \mathbf{J}\rangle.$$

Now,

$$\mathbf{S} \cdot \mathbf{J} = \mathbf{S} \cdot (\mathbf{L} + \mathbf{S}) = \mathbf{L} \cdot \mathbf{S} + \mathbf{S}^2.$$

Also,

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L} \cdot \mathbf{S},$$

so

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}\left(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2\right).$$

Therefore,

$$\langle\mathbf{S} \cdot \mathbf{J}\rangle = \frac{1}{2}\left(\langle\mathbf{J}^2\rangle + \langle\mathbf{S}^2\rangle - \langle\mathbf{L}^2\rangle\right).$$

Using

$$\langle\mathbf{J}^2\rangle = J(J+1), \qquad \langle\mathbf{L}^2\rangle = L(L+1), \qquad \langle\mathbf{S}^2\rangle = S(S+1),$$

we obtain

$$g_J J(J+1) = J(J+1) + \frac{1}{2}\left[J(J+1) + S(S+1) - L(L+1)\right].$$

Hence,

$$\boxed{g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}}.$$

b)

Neglecting the nuclear magnetic moment, the magnetic moment is due only to the electronic angular momentum,

$$\boldsymbol{\mu} = -g_J \mu_B \mathbf{J}.$$

For the total atomic angular momentum

$$\mathbf{F} = \mathbf{I} + \mathbf{J},$$

define

$$\boldsymbol{\mu} = -g_F \mu_B \mathbf{F}.$$

Projecting the magnetic moment onto $\mathbf{F}$ gives

$$g_F \langle \mathbf{F}^2 \rangle = g_J \langle \mathbf{J} \cdot \mathbf{F} \rangle.$$

Since

$$\mathbf{J} \cdot \mathbf{F} = \mathbf{J} \cdot (\mathbf{I} + \mathbf{J}) = \mathbf{I} \cdot \mathbf{J} + \mathbf{J}^2,$$

and

$$\mathbf{F}^2 = \mathbf{I}^2 + \mathbf{J}^2 + 2\mathbf{I} \cdot \mathbf{J},$$

we have

$$\mathbf{J} \cdot \mathbf{F} = \frac{1}{2} \left(\mathbf{F}^2 + \mathbf{J}^2 - \mathbf{I}^2 \right).$$

Using

$$\langle \mathbf{F}^2 \rangle = F(F+1), \qquad \langle \mathbf{J}^2 \rangle = J(J+1), \qquad \langle \mathbf{I}^2 \rangle = I(I+1),$$

gives

$$g_F F(F+1) = \frac{g_J}{2} \left[F(F+1) + J(J+1) - I(I+1) \right].$$

Therefore,

$$\boxed{g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}}.$$

c)

For the term $^2P_{1/2}$,

$$L = 1, \qquad S = \frac{1}{2}, \qquad J = \frac{1}{2}.$$

From part (a),

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} = \frac{2}{3}.$$

Coupling $\mathbf{J}$ to the nuclear spin $\mathbf{I}$ gives

$$F = |I - J|, \ldots, I + J,$$

so for $J = \frac{1}{2}$,

$$F = I + \frac{1}{2}, \qquad F = I - \frac{1}{2}.$$

Using

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)},$$

for $F = I + \frac{1}{2}$,

$$g_F = \frac{2}{3(2I+1)},$$

while for $F = I - \frac{1}{2}$,

$$g_F = -\frac{2}{3(2I+1)}.$$

Thus,

$$\boxed{g_F = \begin{cases} \dfrac{2}{3(2I+1)}, & F = I + \dfrac{1}{2}, \\ -\dfrac{2}{3(2I+1)}, & F = I - \dfrac{1}{2}. \end{cases}}$$

In a weak magnetic field, each hyperfine level splits into $2F + 1$ magnetic sublevels,

$$M_F = -F, -F+1, \ldots, F,$$

with Zeeman shift

$$\boxed{\Delta E_Z = g_F \mu_B M_F B.}$$

Hence the two hyperfine levels have opposite signs of g_F and therefore split in opposite directions with increasing M_F.

Assignment 4

Question 1: Hartree–Fock

The one-electron orbitals are given by

$$\psi_a(\mathbf{r}_i) = \frac{P_{n_a l_a}(r_i)}{r_i} Y_{l_a m_a}(\Omega_i)\chi_{\mu_a}(i),$$

where $a = (n_a, l_a, m_a, \mu_a)$ labels an orbital, and $i = 1, 2$ labels the electron coordinates.

The direct and exchange Coulomb matrix elements are

$$g_{abab} = \left\langle ab \left| \frac{1}{r_{12}} \right| ab \right\rangle, \qquad g_{abba} = \left\langle ab \left| \frac{1}{r_{12}} \right| ba \right\rangle,$$

where $|ab\rangle = \psi_a(1)\psi_b(2)$ and $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$. In general,

$$g_{abcd} = \iint \psi_a^*(1)\psi_b^*(2)\frac{1}{r_{12}}\psi_c(1)\psi_d(2)\, d^3r_1\, d^3r_2.$$

Using the given multipole expansion,

$$\frac{1}{r_{12}} = \sum_{k=0}^{\infty}\sum_{q=-k}^{k} (-1)^q \frac{r_<^k}{r_>^{k+1}} C_{-q}^k(\Omega_1) C_q^k(\Omega_2),$$

where

$$C_q^k(\Omega) = \sqrt{\frac{4\pi}{2k+1}}\, Y_{kq}(\Omega),$$

the radial and angular integrations separate. Defining the radial Coulomb integral

$$R_k(a,b,c,d) = \int_0^\infty \int_0^\infty P_a(r_1)P_c(r_1)\frac{r_<^k}{r_>^{k+1}} \times P_b(r_2)P_d(r_2)\, dr_1\, dr_2,$$

where the arguments denote the radial orbital quantum numbers nl, we obtain the following.

Direct Coulomb integral

Substituting the orbitals and multipole expansion gives

$$g_{abab} = \sum_{kq}(-1)^q R_k(a,b,a,b) \times \langle l_a m_a | C_{-q}^k | l_a m_a\rangle \langle l_b m_b | C_q^k | l_b m_b\rangle.$$

The spin overlaps are unity.

Summing over the filled subshell b, the spin sum contributes a factor of two. For the angular part, the spherical-harmonic addition theorem gives

$$\sum_{m_b=-l_b}^{l_b} |Y_{l_b m_b}(\Omega)|^2 = \frac{2l_b+1}{4\pi}.$$

Consequently,

$$\begin{aligned}
&\sum_{m_b} \langle l_b m_b | C^k_q | l_b m_b \rangle \\
&= \int C^k_q(\Omega) \sum_{m_b} |Y_{l_b m_b}(\Omega)|^2 \, d\Omega \\
&= \frac{2l_b+1}{4\pi} \int C^k_q(\Omega) \, d\Omega \\
&= (2l_b+1)\delta_{k0}\delta_{q0}.
\end{aligned}$$

Thus only the monopole term $k=0$ survives. Since $C^0_0 = 1$,

$$\boxed{\sum_{m_b,\mu_b} g_{abab} = 2(2l_b+1) R_0(n_a l_a, n_b l_b, n_a l_a, n_b l_b).} \tag{3.64}$$

Exchange Coulomb integral

Interchanging the orbital labels in the ket gives

$$\begin{aligned}
g_{abba} = \delta_{\mu_a\mu_b} \sum_{kq} (-1)^q R_k(a,b,b,a) \\
\times \langle l_a m_a | C^k_{-q} | l_b m_b \rangle \\
\times \langle l_b m_b | C^k_q | l_a m_a \rangle.
\end{aligned}$$

The factor $\delta_{\mu_a\mu_b}$ follows from spin orthogonality, so only equal spin projections contribute.

Using the Wigner–Eckart theorem,

$$\begin{aligned}
&\langle l_b m_b | C^k_q | l_a m_a \rangle \\
&\quad = (-1)^{l_b - m_b} \begin{pmatrix} l_b & k & l_a \\ -m_b & q & m_a \end{pmatrix} \langle l_b \| C^k \| l_a \rangle,
\end{aligned}$$

where the reduced matrix element is

$$\begin{aligned}
\langle l_b \| C^k \| l_a \rangle = (-1)^{l_b} \sqrt{(2l_b+1)(2l_a+1)} \\
\times \begin{pmatrix} l_b & k & l_a \\ 0 & 0 & 0 \end{pmatrix}.
\end{aligned}$$

Using $(C^k_q)^\dagger = (-1)^q C^k_{-q}$, the angular products become absolute squares:

$$\begin{aligned}
&\sum_{m_b,q} (-1)^q \langle l_a m_a | C^k_{-q} | l_b m_b \rangle \langle l_b m_b | C^k_q | l_a m_a \rangle \\
&\quad = \sum_{m_b,q} \left| \langle l_b m_b | C^k_q | l_a m_a \rangle \right|^2 \\
&\quad = |\langle l_b \| C^k \| l_a \rangle|^2 \sum_{m_b,q} \begin{pmatrix} l_b & k & l_a \\ -m_b & q & m_a \end{pmatrix}^2 .
\end{aligned}$$

By orthogonality of Wigner $3j$ symbols,

$$\sum_{m_b,q} \begin{pmatrix} l_b & k & l_a \\ -m_b & q & m_a \end{pmatrix}^2 = \frac{1}{2l_a+1}.$$

Combining this with $\sum_{\mu_b} \delta_{\mu_a \mu_b} = 1$ gives

$$\boxed{\begin{aligned}
\sum_{m_b,\mu_b} g_{abba} = \sum_k & \frac{|\langle l_b \| C^k \| l_a \rangle|^2}{2l_a+1} \\
& \times R_k(n_a l_a, n_b l_b, n_b l_b, n_a l_a).
\end{aligned}} \tag{3.66}$$

The permitted multipoles satisfy

$$|l_a - l_b| \le k \le l_a + l_b, \qquad l_a + l_b + k \text{ even}.$$

Question 2: Zeeman Slower

(a) Maximum scattering rate

A Zeeman slower decelerates atoms by repeated absorption of photons from a counter-propagating laser. Each absorbed photon transfers momentum $\hbar k$ opposite to the atomic motion, while spontaneous emission averages to zero momentum transfer.

For a saturated two-level transition, the maximum photon scattering rate is

$$R_{\max} = \frac{\Gamma}{2},$$

where Γ is the spontaneous decay rate. The maximum scattering force and deceleration are therefore

$$F_{\max} = \frac{\hbar k \Gamma}{2}, \qquad \boxed{a_{\max} = \frac{\hbar k \Gamma}{2m}}.$$

This limits how rapidly the atoms can be slowed. In practice, a deceleration $a = \eta a_{\max}$, where $0 < \eta < 1$, is chosen to allow for finite laser intensity and deviations from resonance.

(b) Optimal magnetic field

As the atoms decelerate, their Doppler shift decreases. A spatially varying magnetic field compensates for this through the Zeeman effect, maintaining resonance with the slowing laser.

For atoms moving in the positive z-direction against the laser, the resonance condition is

$$\Delta + kv(z) - \frac{\mu_{\text{eff}} B(z)}{\hbar} = 0,$$

where $\Delta = \omega_L - \omega_0$ is the laser detuning and μ_{eff} is the effective magnetic moment of the transition.

For constant deceleration a,

$$v(z) = \sqrt{v_0^2 - 2az} = v_0 \sqrt{1 - \frac{z}{L_0}}, \qquad L_0 = \frac{v_0^2}{2a}.$$

Substituting into the resonance condition gives

$$B(z) = \frac{\hbar}{\mu_{\text{eff}}} \left[\Delta + k v_0 \sqrt{1 - \frac{z}{L_0}} \right].$$

Choosing the magnetic-field offset such that $B(L_0) = 0$ gives the optimal profile

$$\boxed{B(z) = B_0 \sqrt{1 - \frac{z}{L_0}}, \qquad B_0 = \frac{\hbar k v_0}{\mu_{\text{eff}}}.}$$

The square-root profile compensates continuously for the changing Doppler shift, allowing approximately constant deceleration.

(c) Maximum capture velocity

The maximum initial velocity is limited by the available deceleration and the length L of the slower.

Using

$$v_f^2 = v_i^2 - 2aL,$$

and setting $v_f = 0$, we obtain

$$\boxed{v_{\text{capture}} = \sqrt{2aL}}.$$

Thus, a longer slower or a larger scattering force allows faster atoms to be captured. The magnetic-field range and laser detuning must also be sufficient to bring these atoms into resonance initially.

(d) Alternative magnetic-field profiles

To remain resonant, the atomic velocity must follow

$$v(z) = \frac{\mu_{\text{eff}} B(z)/\hbar - \Delta}{k}.$$

Hence the acceleration required to follow a given magnetic-field profile is

$$\boxed{a_{\text{req}}(z) = v(z)\frac{dv}{dz} = \frac{\mu_{\text{eff}} v(z)}{\hbar k}\frac{dB}{dz}}.$$

For continuous resonant slowing, this acceleration must be negative and must satisfy $|a_{\text{req}}| \leq a_{\max}$.

(b) **Gradual linear decrease.**

The magnetic field decreases more gradually than the ideal square-root profile and reaches zero over a longer distance. The required deceleration can therefore remain below the maximum scattering deceleration. The atoms can reach zero velocity, provided they remain resonant and the slower is sufficiently long.

(c) **Steep linear decrease.**

The field decreases too rapidly initially. For a linear profile reaching zero at L_0, the initial required deceleration is

$$|a_{\text{req}}(0)| = \frac{v_0^2}{L_0} = 2a,$$

where $a = v_0^2/(2L_0)$ is the deceleration required by the ideal profile.

If this exceeds the available scattering deceleration, the atoms cannot follow the field, lose resonance, and fail to reach zero velocity. This is the expected outcome for a slower already operating near its acceleration limit.

(d) **Non-monotonic field.**

The initial decrease slows the atoms, but the subsequent increase in magnetic field requires the resonant velocity to increase. Since the counter-propagating laser produces a decelerating force, the atoms cannot continuously follow this trajectory. They may lose resonance, preventing reliable slowing to zero velocity.

(e) **Oscillating field.**

The rapid field oscillations produce corresponding variations in the atomic detuning. If these variations are sufficiently large, the atoms move out of resonance and the slowing force is reduced.

Small or sufficiently rapid oscillations may be tolerated, allowing the atoms to continue slowing along the underlying square-root profile. Thus, reaching zero velocity depends on the amplitude and spatial scale of the oscillations.

Overall, efficient slowing requires the magnetic field to compensate for the changing Doppler shift without demanding more deceleration than the photon scattering force can provide.

Question 3: Lifetimes of atomic states

The spontaneous emission rate for an electric-dipole (E1) transition from an initial state i to a lower state f is, in atomic units,

$$A_{if} = \frac{4\alpha^3}{3t_{\rm au}}(E_i - E_f)^3 \frac{l_>}{2l_i+1}\,|D_{if}|^2\,,$$

where α is the fine-structure constant, $t_{\rm au} = 2.41888\times10^{-17}\,$s is the atomic unit of time, and $l_> = \max(l_i, l_f)$. The radial dipole matrix element is

$$D_{if} = \int_0^\infty P_f(r)\, r\, P_i(r)\, dr.$$

The factor $l_>/(2l_i+1)$ accounts for the angular integration and summation over magnetic substates. The mean lifetime is

$$\boxed{\tau_i = \frac{1}{\sum_f A_{if}}}.$$

Numerical calculation

The radial wavefunctions and energies were calculated using Mathematica's `NDEigensystem` to solve

$$\left[-\frac{1}{2}\frac{d^2}{dr^2} + \frac{l(l+1)}{2r^2} + V_b(r)\right] P_{nl}(r) = E_{nl} P_{nl}(r),$$

with the parametric sodium potential

$$V_b(r) = -\frac{Z}{r} + \frac{Z-\xi}{r}\left(1 - e^{-r/b}\right),$$

where $Z = 11$, $\xi = 1$, and $b = 0.4072$.

The resulting normalised radial wavefunctions were used to evaluate the dipole integrals numerically. The E1 selection rule $\Delta l = \pm 1$ gives the allowed downward transitions

$$3p \to 3s, \qquad 4s \to 3p, \qquad 3d \to 3p.$$

Below is the code used to perform all calculations.

```

ClearAll["Global`*"];

(* Sodium potential and radial Hamiltonian *)
Z = 11; xi = 1; b = 0.4072;
Vb[r_] := -Z/r + (Z - xi)/r (1 - Exp[-r/b]);
H[l_] := -P''[r]/2 + l (l + 1) P[r]/(2 r^2) + Vb[r] P[r];

```

```
(* Solve radial Schrodinger equation, sorted by energy *)
solve[l_, n_] := SortBy[Transpose@NDEigensystem[
   {H[l], DirichletCondition[P[r] == 0, True]},
   P[r], {r, 0, 50}, n,
   Method -> {
     "Eigensystem" -> {"Arnoldi", "Shift" -> -0.5,
       "MaxIterations" -> 10^5},
     "SpatialDiscretization" -> {"FiniteElement",
       {"MeshOptions" -> {"MaxCellMeasure" -> 0.001}}}
   }], First];

s = solve[0, 2]; p = solve[1, 1]; d = solve[2, 1];

(* Extract valence states *)
{E3s, f3s} = s[[1]]; {E4s, f4s} = s[[2]];
{E3p, f3p} = p[[1]]; {E3d, f3d} = d[[1]];

(* Dipole integrals and spontaneous emission rates *)
Dipole[f_, g_] := NIntegrate[Evaluate[r f g], {r, 0, 50}];

alpha = 1/137.035999084;
tAU = 2.4188843265857*10^-17;

rate[Ei_, Ef_, f_, g_, c_] :=
  4 alpha^3 (Ei - Ef)^3 c Abs[Dipole[f, g]]^2/(3 tAU);

(* Mean lifetimes in nanoseconds *)
Grid[
 Prepend[
  {
   {"3p", 10^9/rate[E3p, E3s, f3p, f3s, 1/3]},
   {"4s", 10^9/rate[E4s, E3p, f4s, f3p, 1]},
   {"3d", 10^9/rate[E3d, E3p, f3d, f3p, 2/5]}
  },
  {"State", "Calculated lifetime (ns)"}
 ],
 Frame -> All
]
```

The calculated mean lifetimes are compared with reference values obtained from NIST transition probabilities in Table 1.

Table 1: Calculated and NIST-derived mean lifetimes of neutral sodium.

State	τ_{calc} (ns)	τ_{NIST} (ns)	Error (%)
$3p$	17.83	16.26	9.7
$4s$	39.22	37.59	4.3
$3d$	20.15	18.47	9.1

The calculated lifetimes agree reasonably well with the NIST reference val-

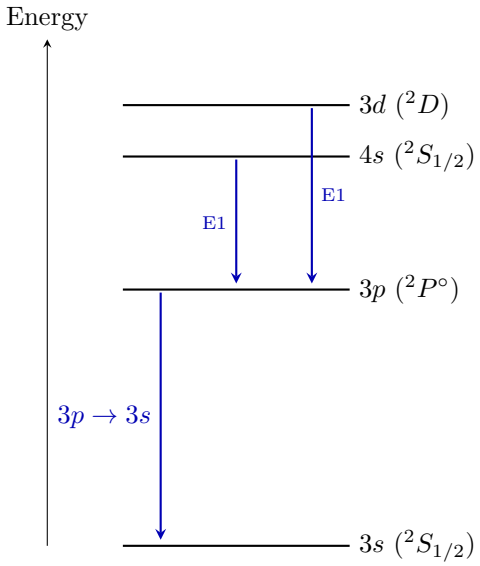


Figure 1: Lowest energy levels of neutral sodium and allowed downward E1 transitions. Fine-structure splitting is omitted for clarity.

ues, with differences of approximately 4–10%. All three calculated lifetimes are slightly longer than the reference values, indicating an underestimate of the spontaneous emission rates.

The main limitation is the use of a single-electron parametric potential to approximate the interaction between the valence electron and the sodium core. This neglects explicit electron correlation and core-polarisation effects. Furthermore, the nonrelativistic model does not resolve fine-structure splitting. Since transition rates depend on both the cube of the transition energy and the square of the dipole matrix element, inaccuracies in the calculated energies and wavefunctions affect the predicted lifetimes.